M1: Pre-Alpha Build

EEL4907 CpE Design 1

Hardened RISC-V Exam Computer

April 19, 2026

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1 Purpose, repository, and design alignment

M1 pre-alpha: architectural skeleton (not a shippable product): how subsystems connect and what exists in-repo today versus the design-draft mockups. **Repository:** `https://github.com/cannon-spencer/riscv-exam-computer`.

Where effort has gone vs. in-repo integration. Most time so far is **hands-on research and bring-up**: SEB and Linux lockdown experiments, distribution and board testing, and tracing the official vs. third-party SEB ecosystem—work that does not yet appear as a single "app" in `exam-env/` but is necessary to choose a viable path. **Custom** kiosk/SEB-wrapper and BEK/CSK integration are **queued for the next implementation phase**; the repository already carries docs/LaTeX, an `exam-env/` placeholder, and the `platform/reptilian-riscv` submodule (**prior** Reptilian/Bananapi-F3 style boot/kernel/partition context). Later milestones may **lean Debian** alongside Linux SEB-class tooling; the final OS line is still open.

SEB upstream references. Official Safe Exam Browser org: `https://github.com/safeexambrowser`. Third-party Linux builds (e.g. v0.4.1): `https://github.com/Jvr2022/seb-linux/releases/tag/v0.4.1`. These inform Linux-on-RISC-V testing; they are not a guarantee of LMS/vendor support compared to official Windows/macOS/iOS SEB.

2 Architectural skeleton (use case: Canvas exam session)

Use case: student authenticates and completes a Canvas exam in a locked browser. Table 1 summarizes how **major subsystem concerns** connect to the planned structure (components, flows, boundaries). This is an architectural snapshot—not a claim that every mechanism is implemented or certified on hardware. The design-draft layered figure remains the visual reference; the table ties it to `exam-env/`, the platform submodule, and HTTPS/IPC.

Subsystem concern	Prepared structure (pre-alpha)
External interface	Kiosk surface: full-screen browser + session shell; navigation mediated by planned Wayland kiosk + browser. Launch reads SEB/exam config and policy from persistent store (`exam-env/` / image path TBD).
Persistent state	Store: SEB XML/plist, exam policy, image/boot metadata; layout tied to `platform/reptilian-riscv/` (partitions, FIT/verity direction). Feeds UI (what is shown) and internals (BEK/CSK, allowlists; SW-01–02, UI-01).
Internal systems	Handling: config parse, kiosk enforcement, browser with BEK/CSK. Reads/writes state only on defined admin or verified paths.
Communication	Channels: HTTPS (Canvas, IdP); local IPC (compositor, session, browser); optional admin UART/SSH. Trust boundaries documented; implementation partial.
Integrity & resilience	Plan: signed boot/FIT, dm-verity-style root, failed-verification handling, recovery via RO root or reimage (upstream notes in submodule). Pre-alpha: architecture only.

Table 1: Subsystem wiring for the Canvas exam-session use case (pre-alpha skeleton).

3 Effort, work completed, and tracking

Phase so far: Substantial time in **research spikes, lab setup, and feasibility testing**: multiple OS and board paths, SEB and lockdown behavior, and scoping the official vs. experimental Linux SEB landscape. An end-to-end Canvas exam on the final RISC-V kiosk image is still ahead; the work below is the foundation for that integration.

Key accomplishments:

- **Arch Linux desktop:** ran Safe Exam Browser to establish lockdown and configuration expectations (reference for a future Linux-native or wrapper path).

- **Linux SEB landscape:** compared official Safe Exam Browser repositories with third-party Linux releases (URLs in Section 1); documented implications for LMS support and maintenance.
- **Banana Pi:** Debian via manufacturer/tutorial; validated boot, storage, and networking on hardware aligned with the design draft.
- **WSL vs. SEB:** characterized policy-relevant behavior on a general-purpose PC; supports requiring a **dedicated** exam device rather than student-owned machines.
- **Repository:** integrated `platform/reptilian-riscv` submodule, documented clone steps, and reserved `exam-env/` for upcoming kiosk/SEB configuration work.

JIRA snapshot (BAC-*):

Stat.	ID	Summary
Done	BAC-1	Design draft: hardware / alternatives
IP	BAC-6,7,10	SEB on Linux; kiosk path; SEB on Orange Pi
Todo	BAC-2	2nd BPi: vendor Linux (tutorial)
Todo	BAC-4,5,8,9	PMP/OpenSBI/secure boot: BPi + Orange Pi + SoC hardening

Table 2: JIRA (BAC) snapshot; full titles in tracker.

Evidence: Git commits (submodule pointer, docs), this report, and dated test notes where applicable; full hour accounting per course policy.

4 Known bugs, limitations, and support friction

Pre-alpha gaps and **support-style friction** (not necessarily one ticket each). Mirror key points in `README.md` if required.

- **Integration depth:** Kiosk/BEK–CSK/compositor wiring is still ahead; current repo content is architecture, submodule context, and placeholders (Section 1).
- **Target E2E:** Full Canvas exam on the locked RISC-V image not yet demonstrated; desk/lab tests do not prove LMS or key handshakes on the target SoC.
- **SEB support model:** Official repos (`https://github.com/safeexambrowser`) center non-Linux; Linux trials (e.g. `https://github.com/Jvr2022/seb-linux/releases/tag/v0.4.1`) are third-party—Canvas/IT compatibility and long-term maintenance are open risks.
- **Submodule vs. Debian:** Reptilian submodule vs. future Debian-first image may force redoing partitions, keys, and secure-boot plumbing; cost unknown.
- **BSP/board noise:** BPi vs. Orange Pi, WiFi/UART, and vendor image quirks burn time as environment/support issues before exam logic.
- **Security not demonstrated:** Secure boot, PMP, verity are goals only; submodule empty clone if `git clone` without `-recurse-submodules` (fix: `git submodule update -init -recursive`).